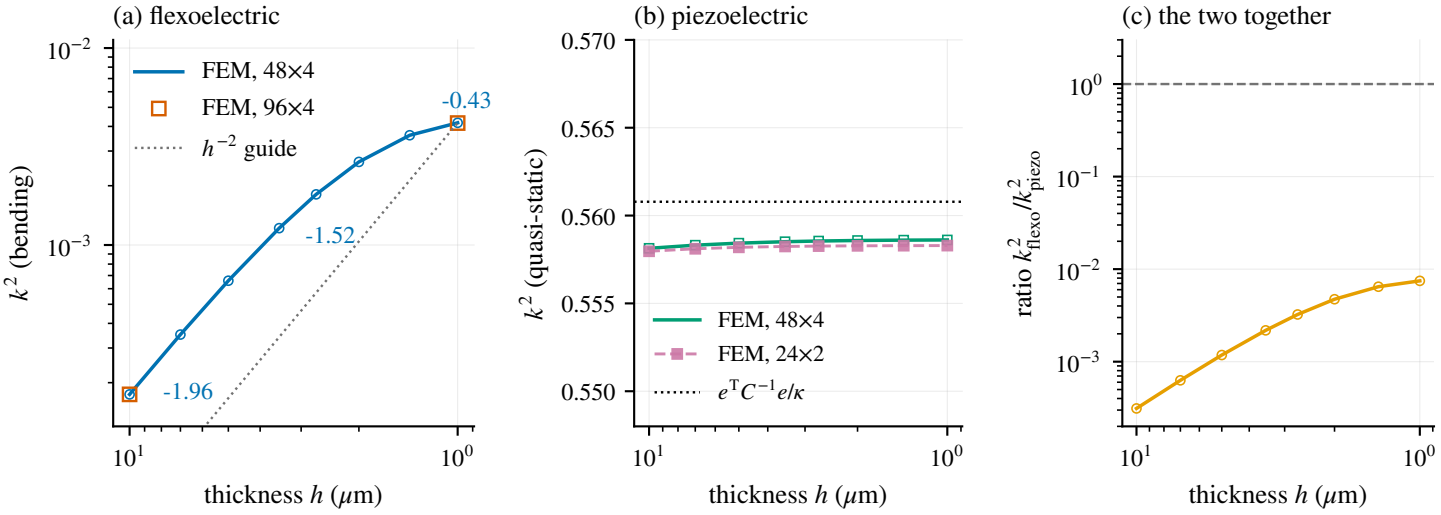
(a) flexoelectric
10^{-2}
10^{-3}
k^2 (bending)
FEM, 48×4
FEM, 96×4
h^{-2} guide
-0.43
-1.52
-1.96
10^1
10^0
thickness h (μm)
(b) piezoelectric
0.570
0.565
0.560
0.555
0.550
k^2 (quasi-static)
FEM, 48×4
FEM, 24×2
$e^{\mathrm{T}}C^{-1}e/\kappa$
10^1
10^0
thickness h (μm)
(c) the two together
10^0
10^{-1}
10^{-2}
10^{-3}
ratio $k^2_{\mathrm{flexo}}/k^2_{\mathrm{piezo}}$
10^1
10^0
thickness h (μm)